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H2**Started on** Monday, 17 August 2026, 10:33 PM**State** Finished**Completed on** Monday, 17 August 2026, 10:37 PM**Time taken** 4 mins 1 sec**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct | Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1 | #include <stdio.h>
2 | int main()
3 | {
4 |     int n, count=0;
5 |     scanf("%d", &n);
6 |     count=2*n+1;
7 |     for(int i=1; i<=n; i++)
8 |     {
9 |         if(n%i==0)
10 |        {
11 |            count++;
12 |        }
13 |    }
14 |    printf("%d", count);
15 |    return 0;
16 | }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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